

Solution to Bayesian Probability Question

Question

An AI-physician was asked about the odds of breast cancer in a woman who was initially thought to have a 2% risk of cancer but received a positive mammogram result. The mammogram accurately classifies 70% of cancerous tumors and 80% of benign tumors. Using Bayes' Rule, calculate the probability.

Solution

Given:

- Prior probability of cancer: $P(C) = 0.02$
- Probability of a positive mammogram given cancer: $P(M^+|C) = 0.70$
- Probability of a positive mammogram given no cancer: $P(M^+|\neg C) = 0.20$

We need to find the probability of cancer given a positive mammogram, $P(C|M^+)$.

Using Bayes' theorem:

$$P(C|M^+) = \frac{P(M^+|C) \cdot P(C)}{P(M^+)}$$

To find $P(M^+)$, we use the law of total probability:

$$P(M^+) = P(M^+|C) \cdot P(C) + P(M^+|\neg C) \cdot P(\neg C)$$

Substituting the given values:

$$P(M^+) = (0.70 \times 0.02) + (0.20 \times 0.98)$$

$$P(M^+) = 0.014 + 0.196 = 0.21$$

Now we substitute back to find $P(C|M^+)$:

$$P(C|M^+) = \frac{0.70 \times 0.02}{0.21} = \frac{0.014}{0.21} \approx 0.0667$$

Answer

The probability of breast cancer given a positive mammogram result is approximately 6.67%.

Question 2

(a) Independent Assumptions

- Burglary and Earthquake are independent.
- JohnCalls is independent of MaryCalls given Alarm.
- JohnCalls and MaryCalls is independent of Burglary or Earthquake given Alarm

(b) Compute $P(+b, +e, +a, +j, -m)$

Given:

- $+b$: Burglary occurred.
- $+e$: Earthquake occurred.
- $+a$: Alarm went off.
- $+j$: John called.
- $-m$: Mary did not call.

We use the dependencies in the Bayesian Network to expand this joint probability:

$$P(+b, +e, +a, +j, -m) = P(+j|+a) \cdot P(-m|+a) \cdot P(+a|+b, +e) \cdot P(+b) \cdot P(+e)$$

Each term can be evaluated from the Conditional Probability Tables (CPTs) if values are provided.

(c) Compute $P(+j, +m)$

We want the probability that both John and Mary called, which requires summing over all possible values of Alarm, Burglary, and Earthquake:

$$P(+j, +m) = P(+j, +m|+a) \cdot P(+a) + P(+j, +m|-a) \cdot P(-a)$$

Breaking it down:

1. Calculate $P(+j, +m|+a)$ as:

$$P(+j, +m|+a) = P(+j|+a) \cdot P(+m|+a)$$

2. Calculate $P(+a)$ by summing over all combinations of Burglary and Earthquake that could trigger the alarm:

$$P(+a) = P(+a \mid +b, +e) \cdot P(+b) \cdot P(+e) + P(+a \mid +b, -e) \cdot P(+b) \cdot P(-e) + P(+a \mid -b, +e) \cdot P(-b) \cdot P(+e) + P(+a \mid -b, -e) \cdot P(-b) \cdot P(-e)$$

3. Calculate $P(-a)$ as:

$$P(-a) = 1 - P(+a)$$

4. Calculate $P(+j, +m \mid -a)$, which is generally a very small probability, depending on any false call rates provided in the CPTs.

Each of these values can be computed directly if CPTs are available in the network.